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PhD candidate in Computer Science at Bar-Ilan University, researching **distance-related graph algorithms**, with papers at **SODA, ICALP, ESA, ISAAC, and DISC**. **Completed my BSc during high school**, followed by an MSc with two SODA papers. Former IDF officer with 5+ years in Unit 8200, where I **built a new cyber R&D team from the ground up**, led critical projects, and recruited for its elite tracks; now researching full-time.

Ramat Gan, Israel

Oct 2023 - Now

- Studying distance-related graph problems. **Distance Oracles, Girth, Diameter, Routing, Fine-Grained complexity.**
- Published 7 papers in graph algorithms; for the full list see avika81.github.io/papers.
- Recipient of the **Clore Scholarship** for exceptional PhD students (2025) and the **President Scholarship** (2024).

Ramat Gan, Israel

Oct 2017 - Mar 2023

- Researched **girth approximations**, published and presented two papers in the prestigious conferences of **SODA'22** and **SODA'23**.
- Collaborated with **leading global experts** in distance-related problems.
- Recipient of the **Nadav Scholarship** for exceptional research, as the sole master's student selected (2022).
- Started During the third year of the BSc, and was a full-time officer at the IDF for most of the degree.

Ramat Gan, Israel

Jul 2015 - Aug 2018

- Graduated with honors at the age of 17.

BIU

Israel

Taiwan

Israel

Germany

Israel

Israel

Tel Aviv, Israel

May 2022 - May 2024

- Formed a new team of Cyber R&D developers to solve crucial problems in the core of the unit's operations.
- Multidisciplinary team operating at cyber security and intelligence interface. Worked in **Python, C, ASM, HTML/JS, and SQL**.
- Mentored new team members, built the team's training program, and led numerous cross-team projects.

Tel Aviv, Israel

Aug 2021 - May 2022

- Oversaw and redesigned the entire recruitment process for the most elite tracks of Unit 8200, conducting over 500 interviews.

Tel Aviv, Israel

Feb 2019 - Aug 2021

- Practical cyber security operations in Windows environments and large networks. Worked mainly in **Python and C++**.
- Developed new systems and optimized existing methods to increase efficiency and achieve outcomes previously unattainable.

U.S.A

May 2018 - Feb 2019

- Applied linear programming to develop and optimize employee scheduling algorithms, collaborating with multi-country teams.

Miscellaneous Git, bash, Docker, $\TeX$, Hebrew (native), English (fluent), French (passable). Interests: chess, bughouse, reading.